Vocalis: Contrastive Speech Analytics via Speaker-Normalized Prosodic Grounding

Track C Technical Report

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1. Executive Summary

Public speaking evaluation tools typically fail in one of two ways: either they rely on black-box LLM wrappers over raw Automatic Speech Recognition (ASR) transcripts—which are blind to pitch, loudness, and sub-word timing—or they apply rigid acoustic thresholds (e.g., raw Hz pitch variance or absolute words per minute) that penalize speakers simply for having a different vocal tract, microphone gain, or baseline speaking style.

Vocalis solves this through **contrastive, speaker-normalized signal processing** anchored to word-level forced alignment. By comparing a participant's delivery word-by-word against a reference ("ideal") recording after normalizing each speaker to their own vocal baseline, Vocalis isolates **local prosodic departures** from **global speaker identity**.

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2. Data Engineering & Contrastive Corpus Construction (30%)

2.1 Hybrid Corpus Design

To evaluate temporal grounding rigorously, a dataset needs both **exact millisecond ground truth** across a controlled severity gradient and **real human-performed speech anomalies** to test cross-speaker generalization:

1. **Public-Domain Reference Baselines (4 Speeches, 371 s total)**:

- **Formal Rhetorical Oratory**: JFK Inaugural Address (`kennedy_inaugural`, 89.4 s, 158 words, about 105 WPM, frequent 1.5–1.7 s rhetorical pauses).

- **Measured Studio Address**: Eisenhower Farewell Address (`eisenhower_farewell`, 87.9 s, 190 words, approximately 130 WPM).

- **Conversational Broadcast Reading**: FDR Fireside Chat 1 (`fdr_fireside_01`, 88.5 s, 259 words, approximately 176 WPM).

- **Interpretive Reading**: Lincoln Gettysburg Address (`lincoln_gettysburg`, 104.0 s, 268 words, approximately 155 WPM).

2. **Programmatic Bad-Mirror Gradient (72 Clips = 4 Speeches × 6 Flaws × 3 Severities)**:

- Candidate flaw windows of 4–9 s are selected automatically on phrase boundaries (`dataset_builder/windows.py`), splitting over-long sentences at their largest natural internal pause.

- **Pitch-Synchronous Overlap-Add (PSOLA)**: Standard phase-vocoder time-stretching (`librosa.effects.time_stretch`) introduces metallic phase smearing that a detector could exploit as a shortcut. Vocalis uses Praat's PSOLA resynthesis (`Lengthen (overlap-add)`), seeded deterministically (`random_initializeWithSeedUnsafelyButPredictably`), to alter duration without altering pitch or introducing phase artifacts.

- **Exact Output-Timeline Ground Truth**: Every perturbation function computes its exact start and end timestamps on the **modified** timeline and is verified independently by `scripts/verify_dataset.py` (72/72 clips verified, 0 failures, bit-identical across processes).

3. **Real Human-Performed Validation Takes (`own_recordings` & `own_recordings_b`)**:

- Four takes recorded by the author on consumer hardware: three takes of `lincoln_gettysburg` (`own_good`, `own_flawed_moderate`, `own_flawed_subtle`) with human-performed pace, monotone, pause, and volume flaws on designated sentences, plus an untouched held-out clean take (`ownb_good`) on `kennedy_inaugural`.

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3. Mathematical Rigor of Feature Extraction & Normalization (20%)

All acoustic features are extracted on a unified $\Delta t = 10$ ms frame grid at $f_s = 16$ kHz:

1. **Two-Pass F_0 Estimation & Semitone Conversion**:

- Pass 1 runs autocorrelation pitch tracking over $[75, 500]$ Hz to determine the speaker's first and third pitch quartiles (Q_1, Q_3).

- Pass 2 restricts search to $[0.75Q_1, 2.5Q_3]$ to suppress octave halving/doubling, followed by iterative octave-jump repair, single-frame glitch removal, and removal of voiced islands < 50 ms.

- Voiced frames are transformed to **semitones relative to the speaker's median**:

$$s(t) = 12 \log_2 \left(\frac{F_0(t)}{\text{median}_t(F_0)} \right)$$

Because a semitone is a logarithmic frequency ratio, a 3-semitone inflection represents the exact same musical interval for a 110 Hz male voice and a 220 Hz female voice.

2. **P90-Normalized RMS Energy**:

- Short-time RMS energy is computed over 25 ms windows and converted to decibels: $E_{\text{dB}}(t) = 20 \log_{10}(\text{RMS}(t) + \varepsilon)$.

- Subtracting the speaker's 90th-percentile voiced speech level L_{90} yields $E_{\text{rel}}(t) = E_{\text{dB}}(t) - L_{90}$, making loudness invariant to microphone gain.

3. **Cepstral Mean & Variance Normalized MFCCs**:

- 13 Mel-Frequency Cepstral Coefficients are standardized per recording: $\hat{C}_k(t) = (C_k(t) - \mu_k)/\sigma_k$.

4. **Adaptive Noise-Floor Silence Threshold**:

- Rather than a fixed dB cutoff, silence is defined adaptively as:

$$\theta_{\text{sil}} = \min\left(\max(L_{\text{noise}} + 10 \text{ dB},\ L_{90} - 30 \text{ dB}),\ L_{90} - 15 \text{ dB}\right)$$

where L_{noise} is the 5th-percentile frame energy. The upper cap at $L_{90} - 15$ dB prevents recordings with zero dead air from misclassifying quiet speech as silence.

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4. Temporal Grounding & Silence-Refined Alignment (25%)

4.1 Discovering and Fixing the WhisperX Silence-Absorption Failure

During evaluation, comparing oracle word timestamps against WhisperX (`wav2vec2`) forced alignment revealed a critical failure mode: when a 2.5 s pause was inserted between *"man"* and *"holds"* in `kennedy_inaugural__pause_severe`, WhisperX assigned **"holds"** the span $[4.921\text{ s}, 7.882\text{ s}]$—absorbing the entire 2.5 s silence into the word duration. Consequently, naive gap detection saw a 0.02 s gap and misclassified the pause as slow speech, while producing **148 false-positive regions** across the synthetic dataset.

Solution (`refine_word_boundaries`): For every aligned word, Vocalis inspects the frame-level energy contour inside $[t_{\text{start}}, t_{\text{end}}]$. If an internal silent run ≥ 0.20 s is present, the word's boundaries are tightened to its longest contiguous voiced span.

- **Impact**: Synthetic false positives dropped from **148 (1.32/min) to 4 (0.04/min)**, and pause detection rose from **75% to 100%** across all severities with a mean IoU of **0.95**.

4.2 Contrastive Neighbourhood Deviation Scoring

Words are paired one-to-one between the baseline and participant. Metrics are computed over a five-word sliding window, $W_i = [i-2, i+2]$.

Local Pace Ratio

$$r_{\text{pace}}(i) = \frac{\sum_{j \in W_i} d_{\text{base}}(j)}{\sum_{j \in W_i} d_{\text{part}}(j)}$$

This ratio is centered by the speaker's global median pace ratio.

Pitch Spread Ratio

$$r_{\text{pitch}}(i) = \frac{\text{P90}(s_{\text{part}}, W_i) - \text{P10}(s_{\text{part}}, W_i)}{\text{P90}(s_{\text{base}}, W_i) - \text{P10}(s_{\text{base}}, W_i)}$$

This ratio is centered by the global median spread ratio.

Loudness Delta

$$\Delta L(i) = L_{\text{part}}(W_i) - L_{\text{base}}(W_i)$$

This difference is centered by the global median level difference.

Punctuation-Aware Pauses

The excess gap is calculated as:

$$\Delta g_i = g_{\text{part}}(i) - g_{\text{base}}(i) \cdot \max\left(1, \frac{1}{r_{\text{local}}}\right)$$

It must exceed **0.50 seconds** mid-clause or **1.00 second** after terminal or clause punctuation (period, question mark, exclamation mark, comma, semicolon, colon, or em dash).

Persistence Rule (`min_region_s = 2.5s`)

Adjacent flagged words are merged and retained only if their baseline span is at least:

$$2.5 \text{ s}$$

This eliminates transient one- or two-word cross-speaker fluctuations.

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5. Causal Explainability & Deterministic Rubric (25%)

Every flagged region outputs a structured causal explanation (`backend/explain.py`) linking the **physical signal measurement**, the **baseline reference value**, the **perceptual effect**, and **actionable coaching advice**:

> **Example (`kennedy_inaugural__pause_severe`)**: `"Unexpected pause: A 2.56 s silence between 'man' and 'holds', where the reference has about 0.06 s. Advice: Keep clauses connected; pause at punctuation rather than mid-phrase."`

The **Rubric Scoring Engine (`backend/scoring_engine.py`)** maps detected regions deterministically to four 0–100 subscores (`pace_control`, `vocal_variety`, `loudness_consistency`, `pauses_fluency`) and an overall score:

$$\text{Penalty}_{\text{prosodic}} = 100 \times \frac{T_{\text{region, base}}}{T_{\text{total, base}}} \times \min\left(3.0, \frac{|\text{metric}|}{|\text{threshold}|}\right)$$

$$\text{Penalty}_{\text{pause}} = 5 \times \min\left(3.0, \frac{\Delta g}{\theta_{\text{pause}}}\right)$$

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6. Validation Methodology, Negative Results & Reproducibility (10%)

1. **Oracle vs. WhisperX Ablation**: Evaluating with exact analytical timestamps (`--alignment oracle`, 85% recall) vs. real WhisperX alignment (`--alignment whisperx`, 85% recall after boundary refinement) proves that silence-aware refinement recovers the full accuracy of oracle timestamps.
2. **Cross-Speech Calibration (Documented Negative Result)**: We implemented 2-fold cross-speech threshold optimization (`evaluation/calibrate.py`), tuning on 2 speeches and evaluating on the held-out 2 speeches. Grid-searched thresholds achieved **0.79 held-out F1** vs. **0.89 held-out F1** for our untuned domain defaults because optimal aggressive thresholds overfit individual speaker cadences. Following strict ML hygiene, we retained the untuned defaults.
3. **Held-Out Cross-Speaker Test**: After freezing all parameters, we recorded an unseen clean take (`ownb_good.wav`) on `kennedy_inaugural`. It produced **1 false positive in 75 s (0.80 FP/min)**, confirming low false-alarm rates on unseen text and an unseen speaker.